Testing Betting Against Beta

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Abstract

The Capital Asset Pricing Model predicts that expected returns are proportional to systematic risk, but the empirical Security Market Line is considerably flatter than this implies. This paper replicates Frazzini and Pedersen (2014) using U.S. equity data from the EODHD database (January 2000 to December 2020). I construct the Betting Against Beta factor and test three propositions: (1) CAPM alpha declines from low-beta to high-beta portfolios, (2) a beta-neutral long-short strategy earns positive risk-adjusted returns, and (3) BAB performance suffers when funding conditions tighten. The BAB factor earns a CAPM alpha of 12.98% per annum ($t = 4.59$), which remains significant under four additional factor specifications. The lagged TED spread is significantly negatively related to BAB returns ($t = -3.52$), linking BAB performance to funding liquidity. At the same time, transaction costs estimated in the literature absorb more than half of the gross premium, suggesting that the anomaly persists because exploiting it is costly.

Keywords: Asset Pricing, CAPM, Beta Anomaly, Leverage Constraints, Factor Investing

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List of Abbreviations

AQR	AQR Capital Management
BAB	Betting Against Beta
CAPM	Capital Asset Pricing Model
CMA	Conservative Minus Aggressive
CRSP	Center for Research in Security Prices
EODHD	End of Day Historical Data
FF3	Fama–French Three-Factor Model
FF5	Fama–French Five-Factor Model
FRED	Federal Reserve Economic Data
HAC	Heteroskedasticity and Autocorrelation Consistent
HML	High Minus Low
LIBOR	London Interbank Offered Rate
NBER	National Bureau of Economic Research
OLS	Ordinary Least Squares
RMW	Robust Minus Weak
SEC	Securities and Exchange Commission
SML	Security Market Line
SMB	Small Minus Big
TED	Treasury-Eurodollar (Spread)
UCITS	Undertakings for Collective Investment in Transferable Securities
UMD	Up Minus Down (Momentum Factor)

1 Introduction

The Capital Asset Pricing Model (Sharpe, 1964; Lintner, 1965) predicts that expected excess returns are proportional to systematic risk, measured by beta. Every asset should lie on the Security Market Line, with alpha equal to zero.

Empirically, this prediction does not hold. Black et al. (1972) documented that the realized Security Market Line is considerably flatter than the CAPM predicts: high-beta stocks earn lower returns than they should, while low-beta stocks earn more. As Fama and French (1992) show for the post-1963 U.S. sample, beta alone does not explain the cross-section of average stock returns.

Why does this anomaly exist, and why does it persist? Frazzini and Pedersen (2014) propose an answer rooted in the portfolio constraints faced by institutional investors, offering an equilibrium rationale for both the anomaly's existence and its persistence. The CAPM assumes investors can borrow freely at the risk-free rate to lever their portfolios. In practice, many institutional investors, including pension funds, mutual funds, and insurance companies, face explicit or implicit restrictions on leverage. Because they cannot amplify returns through borrowing, these constrained investors tilt toward high-beta assets, which offer higher market exposure per dollar invested. This excess demand drives high-beta prices up and expected returns down.

Frazzini and Pedersen (2014) formalize this mechanism in the *Margin-CAPM* and construct the *Betting Against Beta* (BAB) factor: a portfolio that goes long leveraged low-beta assets and short deleveraged high-beta assets, designed to isolate the alpha spread while remaining approximately market-neutral.

The overarching question this paper addresses is: *Why is the empirical Security Market Line flatter than the CAPM implies, which market frictions generate this flattening, and can these distortions be systematically exploited through a Betting Against Beta framework?*

To answer this, I replicate the Frazzini and Pedersen (2014) methodology and test three of their core propositions:

Proposition 1 (Flat Security Market Line). Constrained investors bid up high-beta assets. Consequently, high beta is associated with low CAPM alpha, and the empirical Security Market Line is flatter than theory predicts.

Proposition 2 (Positive BAB Returns). A Betting Against Beta factor that goes long low-beta assets and short high-beta assets, with leverage adjustments to achieve beta neutrality, earns significant positive risk-adjusted returns.

Proposition 3 (Funding Conditions). When funding constraints tighten, proxied by

widening credit spreads, arbitrageurs must reduce leveraged positions. BAB, as a leveraged strategy, suffers contemporaneously.

The main contribution is an independent replication using the EODHD database rather than CRSP, covering a more recent sample period (2000–2020). Because EODHD includes delisted stocks, the sample mitigates survivorship bias. I additionally test Proposition 3 using the TED spread as a proxy for funding conditions. My results confirm the original findings: the BAB factor earns significant positive alpha across five factor models, and its performance weakens during periods of elevated funding stress.

The paper is organized as follows. Section 1.1 clarifies the scope and limitations of this study. Section 2 develops the theoretical foundations, presenting the standard CAPM and the Margin-CAPM extension. Section 3 describes the BAB portfolio construction in detail. Section 4 explains the statistical methodology. Section 5 describes the data and implementation. Section 6 presents the empirical results. Section 7 discusses limitations and robustness. Section 8 concludes.

1.1 Scope and Limitations

The Margin-CAPM of Frazzini and Pedersen (2014) generates five testable propositions. This paper tests the first three: the flat Security Market Line (Proposition 1), positive BAB returns (Proposition 2), and sensitivity to funding conditions (Proposition 3). Frazzini and Pedersen (2014) additionally present cross-asset-class evidence for Propositions 1 and 2 using Treasury bonds, corporate bonds, and equity index futures. This study focuses exclusively on U.S. equities.

Proposition 4 predicts that increased funding liquidity risk compresses cross-sectional beta dispersion toward one. Novy-Marx and Velikov (2022) demonstrate that this compression may reflect estimation artifacts, specifically beta shrinkage from measurement error, rather than a genuine economic effect. Proposition 5 predicts that leverage-constrained investors hold higher-beta portfolios than unconstrained investors. Testing this requires mutual fund holdings or brokerage account data, which are not available for this study. The restriction to Propositions 1–3 follows Sørensen (2015).

Beyond the scope restriction, Novy-Marx and Velikov (2022) raise broader concerns about the BAB construction itself. They argue that estimating betas from correlations rather than regressions is non-standard, that rank-based weighting effectively mimics equal weighting, and that transaction costs absorb much of the gross premium. Section 7 addresses these points in detail.

2 Theoretical Foundations

2.1 The Capital Asset Pricing Model

The CAPM rests on the foundation of mean-variance optimization (see Cochrane, 2005, Chapters 9 and 20). Investors care only about portfolio expected return and variance, meaning they are indifferent between portfolios with identical first two moments. Combined with the assumptions of homogeneous expectations and unlimited borrowing and lending at the risk-free rate R_f, the model yields the following equilibrium result.

Definition 1 (Beta)**.** *The systematic risk of asset i is measured by its beta coefficient:*

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)} \tag{1}$$

where R_i denotes the return on asset i and R_m the return on the market portfolio.

Beta can be decomposed as follows, which is central to the estimation approach:

$$\beta_i = \rho_{i,m} \cdot \frac{\sigma_i}{\sigma_m} \tag{2}$$

where $\rho_{i,m}$ is the correlation between asset i and the market, and σ_i, σ_m are the respective standard deviations. The correlation component captures co-movement with the market, and the volatility ratio scales it to reflect how much larger or smaller the asset's fluctuations are relative to the market as a whole.

The fundamental pricing equation of the CAPM is:

$$E[R_i] - R_f = \beta_i \cdot (E[R_m] - R_f) \tag{3}$$

The left-hand side represents the expected excess return on asset i. The right-hand side decomposes this into the quantity of systematic risk (β_i) multiplied by the price of risk (the market risk premium).

Definition 2 (Alpha)**.** *The alpha of asset i is the deviation of its expected return from the CAPM prediction:*

$$\alpha_i = E[R_i] - R_f - \beta_i \cdot (E[R_m] - R_f) \tag{4}$$

Under the CAPM, $\alpha_i = 0$ for all assets.

2.2 The Empirical Security Market Line

The Security Market Line (SML) provides a graphical representation of the CAPM: plotting expected return against beta, all assets should lie on a line with intercept R_f and slope equal to the market risk premium.

Empirically, however, the SML is flatter. Black et al. (1972) first documented this systematically, finding that the cross-sectional regression:

$$E[R_i] = \gamma_0 + \gamma_1 \cdot \beta_i \tag{5}$$

yields $\gamma_0 > R_f$ and $\gamma_1 < E[R_m] - R_f$. The intercept is too high and the slope too shallow. High-beta assets earn less than the CAPM predicts (negative alpha), while low-beta assets earn more (positive alpha).

What does this mean? Either the CAPM is misspecified, in that beta does not fully capture systematic risk, or some market friction prevents the anomaly from being arbitraged away. The leverage constraint hypothesis offers a friction-based explanation.

2.3 Leverage Constraints and the Flat SML

Tobin (1958) established the separation theorem: in frictionless markets, all investors hold the same tangency portfolio, adjusting risk exposure solely through leverage. An investor seeking a higher expected return simply borrows at R_f to amplify the tangency portfolio.

This separation breaks down when investors face borrowing constraints. Consider an investor whose constraint binds: she desires more risk than the tangency portfolio offers but cannot lever. Her only alternative is to tilt toward high-beta assets, which provide "embedded leverage," that is, higher market exposure per dollar invested.

When many investors face such constraints, aggregate demand for high-beta assets exceeds the frictionless equilibrium level. Prices rise, which drives down expected returns. Low-beta assets, by contrast, attract too little demand and trade below fundamental value. Their expected returns exceed the CAPM prediction. The result is a flatter equilibrium SML.

2.4 The Margin-CAPM

Frazzini and Pedersen (2014) formalize this intuition in the Margin-CAPM. Consider an overlapping generations economy with heterogeneous agents $i = 1, \ldots, I$. Each agent i has wealth W_i and faces a margin constraint m_i that limits leverage.

2.4.1 The Shadow Price of Leverage

Let ψ_i denote the Lagrange multiplier on agent i's margin constraint. This multiplier has a precise economic interpretation: it represents the marginal utility value of an additional dollar of borrowing capacity. When the constraint does not bind, $\psi_i = 0$; the investor can borrow as much as desired, and additional capacity has no value. When binding, $\psi_i > 0$; the investor would willingly pay a premium for the ability to borrow more.

If an investor is constrained, she cannot achieve her optimal portfolio on the efficient frontier. She would happily pay a fee to borrow more money to invest in the tangency portfolio. Since she cannot borrow, this "unmet desire" manifests as a positive shadow price. The magnitude of ψ_i reflects how severely the constraint distorts her portfolio choice.

2.4.2 From Standard CAPM to Margin-CAPM

Define the wealth-weighted average multiplier across all agents:

$$\bar{\psi} = \frac{\sum_i W_i \psi_i}{\sum_i W_i} \tag{6}$$

This aggregate measure captures the economy-wide "tightness" of leverage constraints. If most wealth is held by constrained institutions (pension funds with regulatory restrictions, mutual funds prohibited from borrowing), $\bar{\psi}$ is large. If most wealth is held by unconstrained investors (hedge funds, proprietary trading desks), $\bar{\psi}$ is small.

The modified Security Market Line in the Margin-CAPM is:

$$E[R_i] - R_f = \beta_i \cdot (E[R_m] - R_f) + (1 - \beta_i) \cdot \psi \tag{7}$$

where $\psi > 0$ is an increasing function of $\bar{\psi}$ that captures the premium associated with constrained borrowing. The CAPM alpha implied by this pricing relation is:

$$\alpha_i^{\text{CAPM}} = (1 - \beta_i) \cdot \psi \tag{8}$$

When no constraints bind ($\bar{\psi} = 0$, hence $\psi = 0$), equation (7) reduces to the standard CAPM. As constraints tighten ($\bar{\psi}$ rises), ψ increases. Low-beta assets ($\beta_i < 1$) then earn positive alpha. High-beta assets ($\beta_i > 1$) earn negative alpha. The cross-sectional regression $E[R_i^e] = \gamma_0 + \gamma_1 \beta_i$ has intercept $\gamma_0 = \psi > 0$ and slope $\gamma_1 = E[R_m] - R_f - \psi < E[R_m] - R_f$, which is precisely the empirical pattern documented by Black et al. (1972).

2.4.3 The Equilibrium Mechanism

The Margin-CAPM equilibrium emerges through the following mechanism:

In the standard CAPM, all investors hold the market portfolio (or equivalently, the tangency portfolio). Each investor adjusts risk by levering or de-levering this portfolio. Risk-return preferences are satisfied through borrowing and lending, not through security selection.

In the Margin-CAPM, constrained investors cannot lever. To achieve higher expected returns, they must tilt toward high-beta assets, which deliver more market exposure per dollar invested. This creates excess demand for high-beta securities, pushing their prices up and expected returns down.

Unconstrained investors (those with $\psi_i = 0$) recognize the mispricing. They short overpriced high-beta assets and go long underpriced low-beta assets. But their capital is finite, and they face their own frictions: shorting costs, funding risk, and margin requirements. In equilibrium, some mispricing remains.

2.5 Limits to Arbitrage

Why don't unconstrained arbitrageurs eliminate the mispricing? Shleifer and Vishny (1997) formalize the limits to arbitrage. Several frictions are relevant:

Shorting constraints. Institutional restrictions limit short-selling. Borrowing shares incurs fees, and some securities are simply unavailable to short. This constrains the ability to bet against overpriced high-beta assets.

Funding liquidity risk. Leveraged positions require margin. During market stress, margin requirements increase and financing costs rise. Arbitrageurs may face margin calls, forcing them to close positions at the worst possible time.

Model risk. Beta estimates are noisy. An arbitrageur cannot be certain whether a stock's high beta reflects true systematic risk or estimation error. This uncertainty limits position sizes.

Proposition 3 predicts that arbitrageurs implementing BAB are themselves vulnerable to funding shocks. When credit conditions tighten, they face higher borrowing costs and potential margin calls, forcing position reductions precisely when the opportunity is greatest. This limits arbitrage and allows the anomaly to persist.

3 The Betting Against Beta Strategy

3.1 Core Idea

The BAB factor exploits the flat SML by going long low-beta assets (underpriced, positive alpha) and short high-beta assets (overpriced, negative alpha), with leverage adjustments to target market neutrality.

Definition 3 (BAB Portfolio)**.** *Let securities be ranked by estimated beta in ascending order. Define rank* $z_i = \text{rank}(\beta_i)$ *and average rank* $\bar{z} = n^{-1} \sum_i z_i$*. Assign securities to:*

- *Low-beta portfolio* L*: all securities with* $\beta_i < \text{median}(\beta)$
- *High-beta portfolio* H*: all securities with* $\beta_i \geq \text{median}(\beta)$

3.2 Rank-Based Weighting

Within each portfolio, securities receive rank-based weights following Frazzini and Pedersen (2014):

$$w_i^L = k \cdot (\bar{z} - z_i)^+ \tag{9}$$

$$w_i^H = k \cdot (z_i - \bar{z})^+ \tag{10}$$

where $(x)^+ = \max(x, 0)$ and the normalizing constant ensures weights sum to unity:

$$k = \frac{2}{\sum_i |z_i - \bar{z}|} \tag{11}$$

3.2.1 Why Rank Weighting?

Why use rank-based weights rather than value weights or equal weights? Each scheme has trade-offs:

Value weighting would weight each security by market capitalization. This approach is dominated by mega-cap stocks, which tend to have betas close to one, producing weak beta dispersion and muted alpha spreads (see Novy-Marx and Velikov, 2022).

Equal weighting treats all securities identically, regardless of size. This over-represents micro-cap stocks, which have high transaction costs, low liquidity, and noisy beta estimates. Implementation would be impractical.

Rank weighting provides a middle ground. It preserves the ordering of betas (securities with more extreme betas receive higher weights within their group) while dampening the influence of outliers. Extreme beta estimates, which may reflect estimation noise rather than true systematic risk, receive limited influence.

A practical consideration reinforces this choice: the EODHD database does not provide shares outstanding, so market-capitalization weights are unavailable for this sample.

Example 1 (Rank Weight Calculation)**.** *Consider a simplified universe of five stocks with estimated betas* $\{0.5, 0.7, 1.0, 1.3, 1.5\}$*. The ranks are* $z = \{1, 2, 3, 4, 5\}$ *with average rank* $\bar{z} = 3$*. The rank deviations are* $\{-2, -1, 0, 1, 2\}$*, summing to* $\sum |z_i - \bar{z}| = 6$*. Thus* $k = 2/6 = 1/3$*.*

For the low-beta portfolio: $w_1^L = (1/3) \times 2 = 2/3$ *and* $w_2^L = (1/3) \times 1 = 1/3$*. The lowest-beta stock receives twice the weight of the second-lowest.*

3.3 Leverage Adjustment for Beta Neutrality

The key innovation of BAB is scaling each portfolio to achieve unit beta:

- Scale the low-beta portfolio by $1/\beta_L$
- Scale the high-beta portfolio by $1/\beta_H$

where β_L and β_H are the weighted-average betas of each portfolio:

$$\beta_L = \sum_{i \in L} w_i^L \cdot \beta_i, \qquad \beta_H = \sum_{i \in H} w_i^H \cdot \beta_i \tag{12}$$

After scaling, each leg has an ex ante beta of exactly one. The long-short difference then has ex ante beta of approximately zero, isolating the alpha spread from market exposure.

3.4 Dollar Neutrality versus Beta Neutrality

An important distinction separates BAB from standard long-short factors. Most factor portfolios (such as HML for value or UMD for momentum) are dollar neutral: the long and short legs have equal dollar amounts (e.g., \$100 long, \$100 short). BAB is not dollar neutral. It is beta neutral.

Consider typical portfolio betas. Suppose $\beta_L = 0.6$ and $\beta_H = 1.4$. To achieve unit beta exposure:

$$\text{Long position} = \frac{1}{\beta_L} = \frac{1}{0.6} \approx \$1.67$$
$$\text{Short position} = \frac{1}{\beta_H} = \frac{1}{1.4} \approx \$0.71$$

The investor is long \$1.67 for every \$0.71 sold short, a net long position of approximately \$0.96. This is not a zero-investment strategy in the traditional sense. The investor has substantial capital deployed on the long side.

3.4.1 Self-Financing Structure

Despite the dollar imbalance, BAB is self-financing in the following sense. The return formula (equation 13 below) computes the difference between two excess returns, that is, returns above the risk-free rate. The implicit assumption is that any net cash position earns (or costs) the risk-free rate.

Specifically, the short sale of \$0.71 generates cash proceeds. These proceeds, plus additional borrowing, finance the \$1.67 long position. All cash balances earn or pay R_f. The BAB return captures only the return in excess of this financing cost.

3.4.2 Implications for Funding Risk

The dollar imbalance connects directly to Proposition 3. Because BAB requires significant leverage on the long side, it is exposed to funding conditions: when borrowing costs rise, the cost of maintaining the leveraged long position increases. Section 6.3 tests this empirically.

3.5 Return Calculation

The BAB factor return is:

$$r_t^{\text{BAB}} = \frac{1}{\beta_L}\left(r_t^L - r_t^f\right) - \frac{1}{\beta_H}\left(r_t^H - r_t^f\right) \tag{13}$$

where r_t^L and r_t^H are the portfolio returns, and r_t^f is the risk-free rate.

Expanding this expression:

$$r_t^{\text{BAB}} = \underbrace{\frac{r_t^L - r_t^f}{\beta_L}}_{\text{Leveraged Low-Beta}} - \underbrace{\frac{r_t^H - r_t^f}{\beta_H}}_{\text{Deleveraged High-Beta}} \tag{14}$$

The first term represents the excess return on the low-beta portfolio, scaled up to achieve unit beta. The second term represents the excess return on the high-beta portfolio, scaled down to achieve unit beta. The difference captures the alpha spread while neutralizing market exposure.

Remark 1 (Realized versus Ex Ante Beta)**.** *By construction, BAB has an ex ante beta of zero. Realized betas may differ because ex ante betas are estimated with noise. As Frazzini and Pedersen (2014, p. 10) note, "the BAB factor's realized market loading is not exactly zero, reflecting the fact that our ex ante betas are measured with noise." Section 6.2 reports the realized market exposures.*

4 Statistical Methodology

4.1 Beta Estimation: The Frazzini–Pedersen Approach

Equation (2) decomposes beta into correlation and relative volatility. Frazzini and Pedersen (2014) exploit this decomposition by estimating each component with different time horizons:

- **Volatility:** Rolling one-year window of daily returns (approximately 252 trading days)
- **Correlation:** Rolling five-year window of overlapping three-day returns (approximately 1,260 observations)

The rationale is that volatilities respond quickly to new information, while correlations are more stable. A stock's volatility can double in a month during a crisis: its fundamental co-movement with the market changes more slowly. Using different horizons allows the estimator to capture current volatility conditions while maintaining stable correlation estimates.

4.1.1 Addressing Non-Synchronous Trading

The three-day return is defined as:

$$r_{i,t}^{(3)} = r_{i,t} + r_{i,t-1} + r_{i,t-2} \tag{15}$$

This addresses non-synchronous trading bias, first analyzed by Dimson (1979). The problem arises when securities trade at different times:

Example 2 (Non-Synchronous Trading)**.** *Consider a small-cap stock that trades infrequently. The market crashes at 3:55 PM on Tuesday due to unexpected news. Large-cap stocks immediately reflect the news in their closing prices. However, the small-cap stock's last trade occurred at 2:30 PM, before the news. Its closing price does not reflect the crash.*

Using single-day returns, the correlation between the small stock and the market appears low: the market fell, but the stock did not (on that day). The next morning, the small stock opens sharply lower, "catching up" to the news. The true correlation is high, but single-day measurement misses it.

Summing returns over three days captures these delayed reactions. If the small stock responds to Monday's market news on Tuesday, the three-day return (Monday + Tuesday + Wednesday) for both the stock and market will capture this co-movement.

Remark 2. *Overlapping returns induce mechanical autocorrelation in the correlation estimator because adjacent three-day windows share two days of data. This affects estimation efficiency but does not bias the monthly portfolio returns or their inference, which I address through Newey-West standard errors.*

4.2 Vasicek Shrinkage

Raw beta estimates are noisy, and extreme values tend to overstate true betas due to estimation error. Vasicek (1973) proposed a Bayesian shrinkage estimator that addresses this problem systematically.

4.2.1 The Problem: Estimation Noise and Mean Reversion

Sorting stocks into beta portfolios creates a "winner's curse" problem. The stocks assigned to the "High Beta" bin include two types:

1. Stocks with genuinely high systematic risk
2. Stocks with average systematic risk that received high beta estimates due to sampling noise

The second group's betas will revert toward the mean in subsequent periods, because their high estimates were "lucky draws" that will not persist. This mean reversion causes the high-beta portfolio to have a lower realized beta than its estimated beta, biasing the BAB construction. Shrinkage corrects this problem pre-emptively by pulling extreme estimates toward a prior.

4.2.2 The Shrinkage Formula

The posterior (shrunk) beta estimate is:

$$\beta_i^{\text{shrunk}} = w \cdot \hat{\beta}_i + (1 - w) \cdot \beta_{\text{prior}} \tag{16}$$

where $\hat{\beta}_i$ is the raw estimate, $\beta_{\text{prior}} = 1$ (the market beta), and w is the shrinkage weight.

The optimal shrinkage weight balances two sources of uncertainty:

$$w_t = \frac{\text{Var}(\hat{\beta}) - E[\text{SE}(\hat{\beta})^2]}{\text{Var}(\hat{\beta})} \tag{17}$$

The components are as follows:

$\text{Var}(\hat{\beta})$ is the cross-sectional variance of beta estimates at time t, capturing how spread out the estimates are across all stocks. This reflects both true dispersion in betas and

estimation noise.

$E[\mathrm{SE}(\hat{\beta})^2]$ is the average squared standard error of the estimates. This reflects pure estimation uncertainty.

The difference $\mathrm{Var}(\hat{\beta}) - E[\mathrm{SE}^2]$ estimates the true cross-sectional variance of betas, stripping out the noise component.

The ratio w can be interpreted as:

$$w = \frac{\text{True signal variance}}{\text{Total observed variance}} = 1 - \frac{\text{Noise variance}}{\text{Total variance}} \tag{18}$$

When estimation noise is high relative to true dispersion, w is low, the estimator pulls aggressively toward the prior. As true dispersion increases and estimates become more precise, w approaches one.

Frazzini and Pedersen (2014) adopt a fixed shrinkage weight of $w = 0.6$, placing 40% weight on the cross-sectional prior of $\beta = 1$. This fixed calibration avoids the instability that arises from estimating w_t month by month (where small-sample noise in $\mathrm{Var}(\hat{\beta})$ can produce erratic weights) while still correcting the winner's curse in portfolio sorts. I follow the same fixed-weight specification:

$$\beta_i^{\text{shrunk}} = 0.6 \cdot \hat{\beta}_i^{\text{TS}} + 0.4 \cdot 1.0 \tag{19}$$

A stock with $\hat{\beta}^{\text{TS}} = 2.5$ is thus shrunk to $0.6 \times 2.5 + 0.4 = 1.90$, reflecting that some of the extreme estimate likely captures noise rather than true systematic risk.

4.3 Newey-West Standard Errors

Asset returns exhibit autocorrelation and heteroskedasticity, violating the assumptions underlying ordinary least squares inference.

4.3.1 The Problem

Standard OLS assumes that regression errors are independent and identically distributed (i.i.d.). Financial data violates both assumptions:

Heteroskedasticity. Return volatility varies over time, as calm periods alternate with turbulent ones. Errors are not identically distributed because their variance changes.

Autocorrelation. Returns exhibit serial dependence. Volatility clustering means that large returns (positive or negative) tend to follow large returns. In monthly data, overlapping return windows and microstructure effects create additional serial dependence.

When these assumptions fail, OLS coefficient estimates remain unbiased, but standard errors are incorrect, typically too small. This leads to overstated t-statistics and spurious significance.

4.3.2 The Newey-West Solution

Newey and West (1987) developed a heteroskedasticity and autocorrelation consistent (HAC) covariance estimator. The intuition follows the "sandwich" analogy:

Standard OLS variance is $(X'X)^{-1}\sigma^2$, essentially "bread" on both sides of a homoskedastic variance term.

The Newey-West estimator replaces the middle term with a matrix $\hat{\Omega}$ that accounts for the correlation structure of errors:

$$\hat{V}_{NW} = (X'X)^{-1}\,\hat{\Omega}\,(X'X)^{-1} \tag{20}$$

The matrix $\hat{\Omega}$ incorporates not just contemporaneous error variances but also lagged covariances up to lag L:

$$\hat{\Omega} = \hat{\Gamma}_0 + \sum_{j=1}^{L}\left(1 - \frac{j}{L+1}\right)\left(\hat{\Gamma}_j + \hat{\Gamma}_j'\right) \tag{21}$$

where $\hat{\Gamma}_j$ captures the covariance between errors j periods apart. The Bartlett kernel weights $(1 - j/(L+1))$ ensure positive semi-definiteness while down-weighting distant lags.

4.3.3 Lag Selection

Following Frazzini and Pedersen (2014), I use $L = 6$ lags for monthly data. Newey and West (1994) recommend a data-driven lag length of $L \approx 0.75 \cdot T^{1/3}$, which for $T = 252$ monthly observations yields $L \approx 4.7$; I round up to 6 to be conservative, consistent with the convention in the BAB literature.

4.4 Return Winsorization

I winsorize monthly stock returns cross-sectionally at the 0.5th and 99.5th percentiles before portfolio formation, following Bali et al. (2016), who apply this threshold in their textbook treatment of cross-sectional asset pricing. This mitigates the influence of extreme observations (data errors, corporate events, or lottery-stock effects) on equal-weighted portfolio returns. The procedure caps, rather than removes, extreme values, preserving the full sample size. As I use a commercial data provider (EODHD) rather than the

curated CRSP database employed by Frazzini and Pedersen (2014), this step helps ensure that portfolio returns are not driven by a small number of outliers.

5 Data and Implementation

5.1 Data Sources

Stock price data are obtained from the EODHD API (End of Day Historical Data, All World plan), which provides daily adjusted closing prices for all U.S.-listed securities including delisted stocks. The database covers 21,353 tickers across NYSE, AMEX, and NASDAQ, of which 21,298 (99.7%) were successfully downloaded. After restricting to common stocks with at least one valid monthly return observation, 14,016 tickers remain and form the initial panel of 1,725,527 stock-months. The inclusion of delisted common stocks mitigates survivorship bias (Shumway, 1997). However, EODHD does not provide explicit delisting returns. When a stock delists, its last available adjusted closing price is treated as the final observation, and no further return is computed. Shumway (1997) shows that ignoring delisting returns biases average returns upward, particularly for high-beta stocks that delist due to financial distress. This bias works against my findings: it inflates D10 returns and thus understates the D1–D10 alpha spread, making my estimates of BAB performance conservative. Daily prices are adjusted for stock splits and dividends by the data provider.

EODHD differs from the CRSP database used by Frazzini and Pedersen (2014) in coverage and quality controls. CRSP access requires an institutional license that was not available for this thesis. The university's Refinitiv Datastream terminal can retrieve individual security histories, but the hardware and software limitations of the available workstation made downloading the full U.S. equity universe impractical within the scope of this project. EODHD provides the necessary security-level coverage at the cost of weaker quality controls relative to CRSP. The winsorization procedure described in Section 4.4 addresses remaining data quality differences.

Market returns and the risk-free rate come from the Kenneth French Data Library. I also obtain Fama–French three-factor returns (Fama and French, 1993), the momentum factor (Carhart, 1997), Fama–French five-factor returns (Fama and French, 2015), and Hou–Xue–Zhang q-factor returns (Hou et al., 2015) for multi-factor regressions. The TED spread, defined as the difference between the three-month LIBOR rate and the three-month Treasury bill rate, is obtained from the Federal Reserve Bank of St. Louis (FRED). As an external benchmark, I use the published BAB factor returns from AQR Capital Management (2024).

5.2 Sample Filters

Securities enter the sample when they satisfy five criteria: (i) median monthly closing price at or above \$5, excluding penny stocks subject to extreme microstructure noise; (ii) absolute estimated beta at or below 5, removing estimation artifacts. A stock with $|\hat{\beta}| > 5$ implies movement of more than five times the market, which for individual equities is virtually always an artifact of thin trading or data errors rather than genuine systematic risk; (iii) listing on a major U.S. exchange (NYSE, AMEX, or NASDAQ); (iv) mean daily trading volume of at least 1,000 shares, establishing a minimal liquidity floor; and (v) at least 36 months of panel history, ensuring stable beta estimates. These filters reduce the initial panel of 1,725,527 stock-months across 14,016 tickers to 984,913 stock-months across 7,946 tickers (57.1% retained).

The \$5 price filter partially addresses concerns raised by Novy-Marx and Velikov (2022), who show that BAB's apparent alpha is concentrated in micro-cap stocks with high idiosyncratic volatility. The history requirement creates a sample selection concern: requiring 36 months of data systematically excludes young firms, which tend to be volatile growth stocks with higher betas. While the inclusion of delisted stocks limits delisting bias, this requirement may underrepresent high-beta securities.

5.3 Sample Period

The sample period spans January 2000 to December 2020 (252 months). The start date reflects the stabilization of EODHD coverage for U.S. equities: the number of active tickers approximately doubles between 1997 and 2000 as the provider's historical database matures. Starting in 2000 avoids potential survivorship artifacts from incomplete early coverage. The end date of December 2020 includes the COVID-19 market disruption, a severe funding liquidity event that provides a natural experiment for testing Proposition 3.

5.4 Timing Convention

Betas are estimated from the trailing windows described in Section 4.1 and sampled at the end of each calendar month. Frazzini and Pedersen (2014) describe sorting "on the basis of their estimated beta at the end of the previous month." In my implementation, the end-of-month beta assigns stocks to deciles for that same month's return. The distinction is economically negligible because the estimation windows (252 trading days for volatility, 1,260 for correlation) make month-to-month beta variation minimal by construction. The beta transition analysis in Section 7.1 confirms this: 86% to 96% of stocks remain in the same beta quintile from one month to the next. All price data are split- and dividend-adjusted closing prices.

6 Empirical Results

With the theoretical framework, portfolio construction, and data in place, I now test the three propositions derived from the Margin-CAPM. Each addresses one dimension of the research question posed in the introduction: whether the SML is indeed flat (Section 6.1), whether the resulting mispricing can be captured by BAB (Section 6.2), and how funding conditions affect the strategy's performance (Section 6.3).

6.1 Test of Proposition 1: The Flat Security Market Line

To test Proposition 1, I sort securities into beta deciles each month and compute CAPM alphas for each decile portfolio, following the standard time-series testing approach for asset pricing models (Cochrane, 2005, Chapter 12). The CAPM regression is:

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_p \left(R_{m,t} - R_{f,t}\right) + \varepsilon_{p,t} \tag{22}$$

with Newey–West standard errors (six lags).

Under the CAPM, all alphas should equal zero. Under the leverage constraint hypothesis, alphas should decline from low-beta to high-beta deciles: positive for decile 1, negative for decile 10.

Table 1 presents the results. The estimated alphas generally decline from the lowest-beta decile (D1) through the highest-beta decile (D10), consistent with the leverage constraint hypothesis. D1 earns a CAPM alpha of 0.69% per month ($t = 4.34$), statistically significant at the 1% level. D10 earns an economically and statistically insignificant alpha of 0.07% ($t = 0.19$). The D1–D10 alpha spread is 0.63% per month, or 7.52% per annum.

Table 1: Decile Portfolio CAPM Alphas

Decile	$\bar{\beta}^{\text{assigned}}$	$\hat{\beta}_{\text{CAPM}}$	$\bar{r}^e$ (%)	$\hat{\alpha}$ (%)	$t(\hat{\alpha})$	R^2
D1 (Low)	0.471	0.356	0.89	0.69	4.34***	0.462
D2	0.646	0.538	1.04	0.75	4.65***	0.620
D3	0.749	0.646	1.04	0.69	3.89***	0.649
D4	0.826	0.759	1.00	0.58	3.12***	0.686
D5	0.894	0.874	0.98	0.50	2.39**	0.711
D6	0.962	0.968	0.98	0.45	1.94*	0.698
D7	1.034	1.073	0.86	0.27	1.17	0.732
D8	1.120	1.213	0.94	0.27	1.09	0.742
D9	1.242	1.439	0.94	0.14	0.56	0.758
D10 (High)	1.574	1.940	1.14	0.07	0.19	0.730
D1–D10				0.63	(7.52% p.a.)	

Notes. Equal-weighted decile portfolios, January 2000 to December 2020 (252 months). Newey–West t-statistics (6 lags). ***/**/*: 1%/5%/10%.

Frazzini and Pedersen (2014, Table 3, p. 13) report CAPM alphas of 0.52% (P1) and −0.10% (P10) per month, a D1–D10 spread of 0.62%. My decile results differ from theirs in two respects, both attributable to the equal weighting within deciles that the absence of market capitalization data in EODHD necessitates. First, D2 alpha (0.75%, $t = 4.65$) marginally exceeds D1 alpha (0.69%, $t = 4.34$), a difference of 0.06% per month that is economically negligible and statistically insignificant. Second, D10 alpha is near zero (0.07%, $t = 0.19$) rather than negative as in the original study. Both patterns reflect the same mechanism: equal weighting gives small-cap stocks with noisy returns the same influence as large-cap stocks. In D1, illiquid micro-caps carry a small liquidity discount that slightly depresses the average return. In D10, lottery-type stocks with occasional extreme positive returns inflate the average, masking the negative alpha that emerges under value weighting. Equal weighting shifts both endpoints upward, but the spread is preserved: my D1–D10 CAPM alpha spread of 0.63% per month matches the 0.62% in the original study to within one basis point. The 7.52% per annum spread confirms a Security Market Line that is flatter than the CAPM predicts. The BAB factor itself uses rank-based weights and is not affected by this limitation (Section 7.2).

Figure 1 presents the empirical Security Market Line. Panel (a) plots the assigned portfolio beta against mean excess return for each decile. The theoretical SML, which passes through the origin with a slope equal to the market risk premium, substantially overstates expected returns for high-beta portfolios. The empirical SML, fitted by OLS to the ten decile points, is much flatter. Panel (b) displays the generally declining pattern of CAPM alpha across deciles.

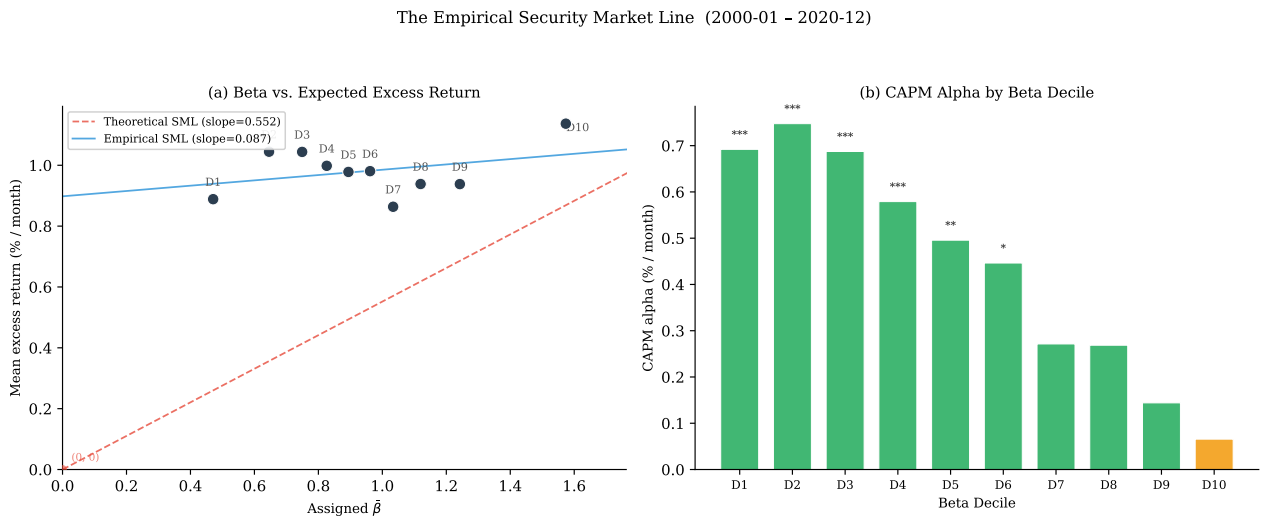

Figure 1: The Empirical Security Market Line. Panel (a) plots the mean monthly excess return against the assigned portfolio beta for ten beta-sorted deciles. The dashed line is the theoretical SML; the solid line is the OLS fit to the decile points. Panel (b) shows CAPM alpha by decile with 95% confidence intervals. Sample: January 2000 to December 2020.

6.2 Test of Proposition 2: BAB Performance

Table 2 summarizes the BAB factor. Over the 252-month sample period, BAB earns a mean return of 0.84% per month (10.11% annualized) with a monthly standard deviation of 3.65% (12.64% annualized), yielding an annualized Sharpe ratio of 0.800. The Sortino ratio, which penalizes only downside deviation, is 0.779, marginally below the Sharpe ratio, reflecting the slight negative skewness of the return distribution. The Calmar ratio (annualized return divided by maximum drawdown) is 0.395. The maximum drawdown of −25.59% occurs during the 2008–2009 financial crisis. The distribution is approximately symmetric (skewness −0.09) with excess kurtosis of 3.02, indicating heavier tails than the normal distribution. BAB is positive in 64.3% of months. The average portfolio betas are $\bar{\beta}_L = 0.63$ and $\bar{\beta}_H = 1.31$, with an average of 2,825 stocks per month.

Table 2: BAB Factor Summary Statistics

Panel A: Return Statistics			
Months	252	Mean (annual)	10.11%
Period	2000-01 to 2020-12	Std (annual)	12.64%
Mean (monthly)	0.84%	Sharpe ratio (ann.)	0.800
Std (monthly)	3.65%	Sortino ratio (ann.)	0.779
Max drawdown	−25.59%	Calmar ratio	0.395
% positive months	64.3%		
Panel B: Distribution			
Skewness	−0.094	Min	−12.75%
Excess kurtosis	3.024	Max	15.03%
Panel C: Portfolio Characteristics			
Avg β_L	0.631	Avg β_H	1.309
Avg stocks/month	2,825		

Notes. The BAB factor was constructed following Frazzini and Pedersen (2014) with rank-based weights and leverage adjustment. Returns in excess of R_f. January 2000 to December 2020.

I test whether BAB earns statistically significant alpha under five-factor model specifications of increasing richness (Cochrane, 2005, Chapter 13). The CAPM regression is:

$$r_t^{\text{BAB}} = a + b \cdot MKT_t + \varepsilon_t \tag{23}$$

I augment this with the Fama–French three-factor model (Fama and French, 1993) (Eq. (24)), the Carhart (1997) four-factor model adding momentum (Eq. (25)), the Hou et al. (2015) q-factor model (Eq. (26)), and the Fama and French (2015) five-factor model (Eq. (27)):

$$r_t^{\text{BAB}} = a + b \cdot MKT_t + s \cdot SMB_t + h \cdot HML_t + \varepsilon_t \tag{24}$$

$$r_t^{\text{BAB}} = a + b \cdot MKT_t + s \cdot SMB_t + h \cdot HML_t + u \cdot UMD_t + \varepsilon_t \tag{25}$$

$$r_t^{\text{BAB}} = a + \beta_m \cdot MKT_t + \beta_{me} \cdot R_{ME,t} + \beta_{ia} \cdot R_{IA,t} + \beta_{roe} \cdot R_{ROE,t} + \varepsilon_t \tag{26}$$

$$r_t^{\text{BAB}} = a + b \cdot MKT_t + s \cdot SMB_t + h \cdot HML_t + r \cdot RMW_t + c \cdot CMA_t + \varepsilon_t \tag{27}$$

The momentum factor (UMD) is the return difference between portfolios of past winners and past losers. As the most robust cross-sectional return predictor (Fama and French, 2008), it presents a natural control for BAB, since low-beta stocks may systematically differ from high-beta stocks in past performance.

Table 3 reports the results. The CAPM specification yields an alpha of 1.08% per month ($t = 4.59$), or 12.98% per annum, significant at the 1% level. The market loading is $\hat{b} = -0.43$. Although BAB is dollar-long (Section 3.4), this does not imply positive market beta: the long dollars are invested in low-beta stocks and the short dollars in high-beta stocks, so the relevant quantity is dollar exposure *times* per-dollar beta. The negative loading arises because Vasicek shrinkage compresses assigned betas toward one. Consider a stock with true $\beta = 2.5$: shrinkage yields $0.6 \times 2.5 + 0.4 = 1.90$, pulling the estimate inward. Applied to the entire portfolio, this inflates $\beta_L^{assigned}$ above its true value and deflates $\beta_H^{assigned}$ below its true value. The long leg is then scaled by $1/\beta_L$ that is too small (under-leveraged), while the short leg is scaled by $1/\beta_H$ that is too large (retaining more market exposure than intended). The net effect is a residual short-market tilt. Frazzini and Pedersen (2014) report a comparable loading of -0.33 for their BAB factor and note that "the BAB factor's realized market loading is not exactly zero, reflecting the fact that our ex ante betas are measured with noise."

The three-factor alpha is 1.09% per month ($t = 5.16$, 13.11% annualized), indicating that the BAB premium is not explained by size (SMB) or value (HML) factors. The Carhart four-factor model absorbs part of the alpha through the momentum factor: the estimate falls to 0.98% per month ($t = 4.73$, 11.78% annualized), reducing the unexplained premium by approximately 1.3 percentage points. The significant positive UMD loading

($\hat{u} = 0.27$, $t = 5.45$) indicates that BAB covaries positively with momentum returns, consistent with low-beta stocks overlapping with past winners, high-beta stocks overlapping with past losers, or both. Nevertheless, the alpha remains highly significant. The q-factor model, which includes profitability and investment factors, reduces the alpha further: the estimate falls to 0.83% per month ($t = 4.08$, 9.99% annualized). The five-factor model yields an alpha of 0.79% per month ($t = 3.48$, 9.45% annualized).

Table 3: BAB Factor Regressions

	CAPM Eq. (23)	FF3 Eq. (24)	Carhart Eq. (25)	q-factor Eq. (26)	FF5 Eq. (27)
$\hat{\alpha}$ (%/mo)	1.08	1.09	0.98	0.83	0.79
$t(\hat{\alpha})$	4.59***	5.16***	4.73***	4.08***	3.48***
$\hat{\alpha}$ (%/yr)	12.98	13.11	11.78	9.99	9.45
$\hat{b}$	−0.434	−0.404	−0.273	−0.252	−0.297
	(−5.21)	(−4.79)	(−3.61)	(−3.27)	(−3.57)
$\hat{s}$		−0.152	−0.203		−0.010
		(−2.07)	(−3.09)		(−0.09)
$\hat{h}$		0.123	0.210		−0.126
		(0.97)	(2.03)		(−1.42)
$\hat{u}$			0.267		
			(5.45)		
$\hat{\beta}_{R_{ME}}$				0.006	
				(0.07)	
$\hat{\beta}_{R_{IA}}$				0.281	
				(1.97)	
$\hat{\beta}_{R_{ROE}}$				0.408	
				(5.27)	
$\hat{r}$					0.371
					(3.18)
$\hat{c}$					0.397
					(2.07)
R^2	0.295	0.328	0.448	0.409	0.393
N	252	252	252	252	252

Notes. Newey–West t-statistics (6 lags). ***/**/*: 1%/5%/10% significance. January 2000 to December 2020.

BAB alpha remains statistically and economically significant across all five factor specifications. Even the most demanding model (FF5) leaves nearly 10% per annum unaccounted for, a premium that standard risk factors cannot explain. As a benchmark comparison,

the published AQR BAB factor for U.S. equities over the same period earns 10.09% annualized with a Sharpe ratio of 0.72. The correlation between my replication and the AQR factor is 0.82, confirming that my replication tracks the published series closely.

Figure 2 displays the cumulative growth of $1 invested in the BAB factor versus the market portfolio. BAB outperforms the market on a risk-adjusted basis; the 2008–2009 drawdown of −25.59% remains shallower than the concurrent market decline.

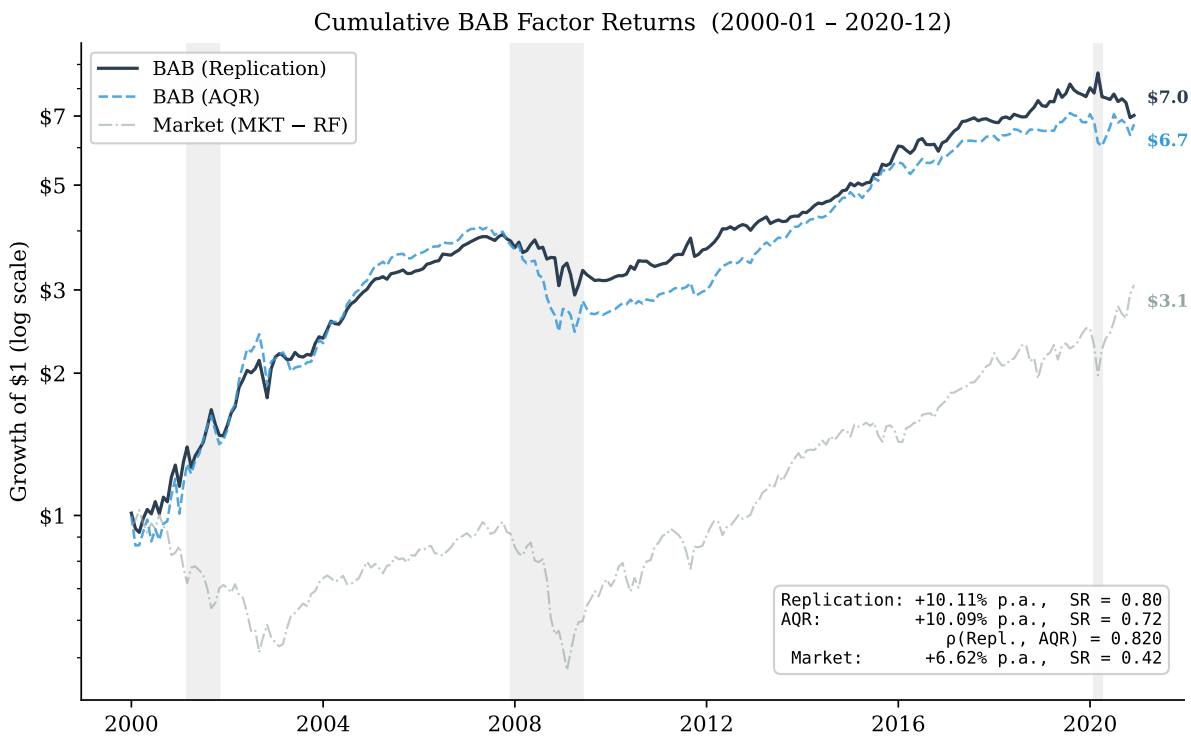

Figure 2: Cumulative Growth of $1 Invested in BAB versus the Market Portfolio and the AQR Benchmark. The figure plots the cumulative value of $1 invested in the BAB replication (solid), the published AQR BAB factor (dashed), and the market excess return (dotted) from January 2000 to December 2020. Grey shading indicates NBER recession periods.

Figure 3 confirms that the declining alpha pattern of Proposition 1 holds across all five factor models. Low-beta deciles retain significant positive alphas throughout; significance fades toward the top deciles and is absent in D9 and D10 under every specification.

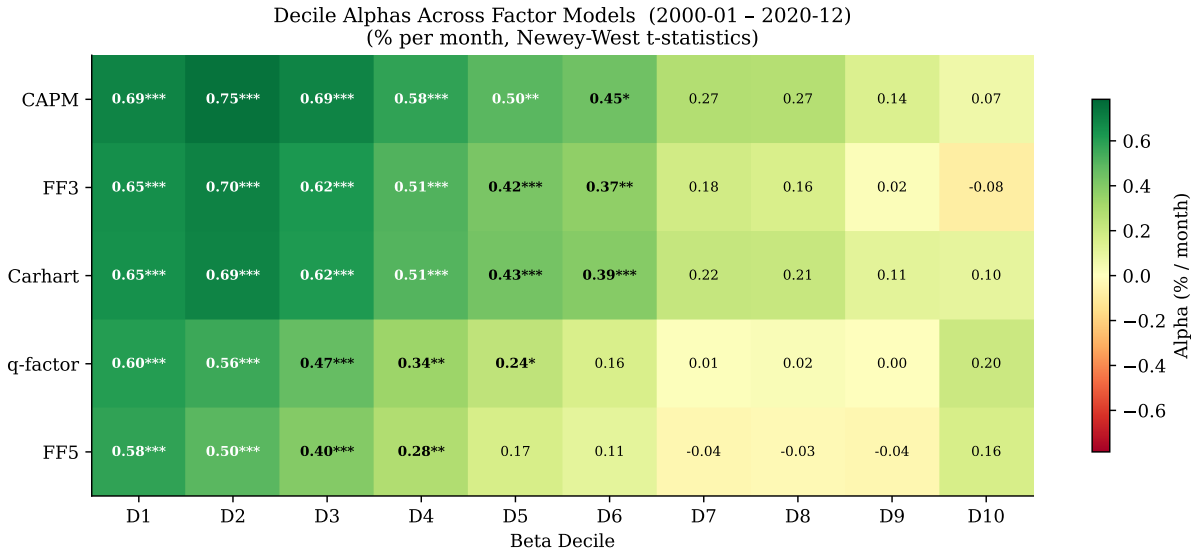

Figure 3: Decile Alphas Across Factor Models. Each cell reports the monthly alpha (%) for a beta decile under the specified model. Significance stars indicate Newey–West t-statistics: ***/**/* at 1%/5%/10%. Sample: January 2000 to December 2020.

Figure 4 disaggregates BAB performance into four sub-periods. The strategy performs strongly during the pre-crisis (2000–2006) and recovery (2009–2015) periods but suffers during the financial crisis (2007–2009), consistent with Proposition 3. Performance deteriorates again in the late sample (2016–2020), with an annualized return of 4.4% and a Sharpe ratio of 0.41, a pattern consistent with post-publication decay (McLean and Pontiff, 2016). Two factors may have contributed. First, low-volatility strategies became increasingly popular after 2014, attracting capital that likely narrowed the alpha spread. Second, the prolonged low-interest-rate environment after 2008 eased borrowing conditions for many investors, weakening the leverage constraint mechanism that the Margin-CAPM describes. I cannot test these explanations with the available data, but both are consistent with a smaller BAB premium when funding constraints are loose.

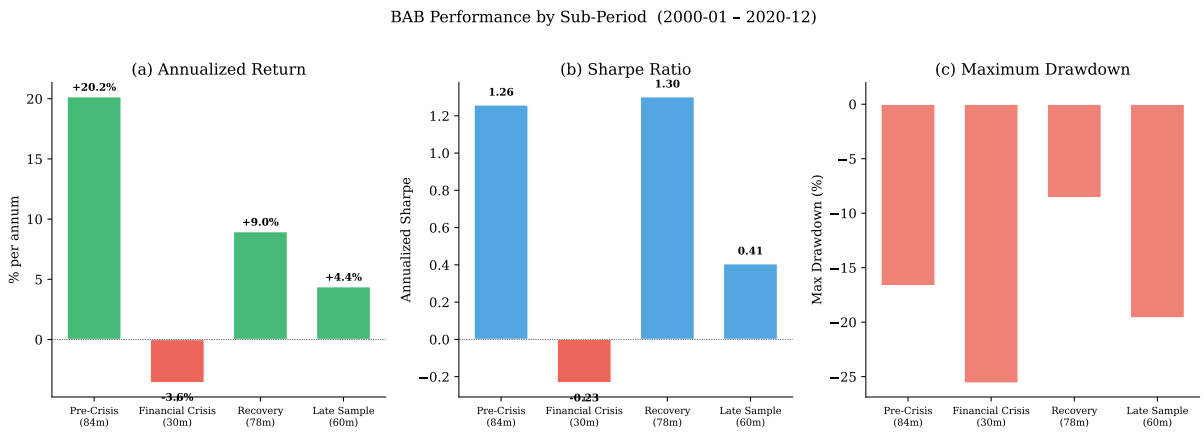

Figure 4: BAB Performance by Sub-Period. Panel (a) shows annualized returns, panel (b) annualized Sharpe ratios, and panel (c) maximum drawdowns for four sub-periods.

6.3 Test of Proposition 3: Funding Conditions

To test Proposition 3, I examine the relationship between BAB returns and the TED spread. Higher TED spreads indicate elevated credit risk and tighter funding. The regression specification follows Frazzini and Pedersen (2014):

$$r_t^{\text{BAB}} = a + b_1 \cdot \Delta TED_t + b_2 \cdot TED_{t-1} + \varepsilon_t \tag{28}$$

where $\Delta TED_t = TED_t - TED_{t-1}$ captures contemporaneous changes.

The prediction $b_1 < 0$ follows from the nature of BAB as a leveraged strategy. When funding costs rise:

1. Arbitrageurs face higher borrowing costs to maintain the leveraged long position.
2. Some arbitrageurs face margin calls, forcing position liquidation.
3. Selling pressure on low-beta stocks (the long leg) depresses their prices.
4. BAB returns fall contemporaneously.

Forced unwinding concentrates losses precisely when funding conditions are tightest, making BAB risk partly endogenous. The lagged level TED_{t-1} captures a separate channel: persistently elevated funding costs may suppress expected returns in subsequent months as arbitrageurs remain constrained.

Table 4 reports the results. Both coefficients carry the predicted negative sign. The contemporaneous change coefficient $\hat{b}_1 = -0.014$ ($t = -1.23$) is not statistically significant at conventional levels. The lagged level coefficient $\hat{b}_2 = -0.014$ ($t = -3.52$) is significant at the 1% level. The regression R^2 is 2.2%, low but typical for single-predictor time-series regressions in asset pricing (Cochrane, 2005, Chapter 13).

Table 4: BAB Returns and the TED Spread

	Coefficient	t-statistic
Intercept ($\hat{a}$)	0.0144	5.61***
ΔTED_t ($\hat{b}_1$)	−0.0137	−1.23
TED_{t-1} ($\hat{b}_2$)	−0.0141	−3.52***
R^2	0.022	
N	252	

Notes. TED spread = 3-month LIBOR minus 3-month T-bill rate (percentage points). Newey–West t-statistics (6 lags). ***/**/*: 1%/5%/10% significance. January 2000 to December 2020.

The insignificance of $\hat{b}_1$ reflects how institutional arbitrageurs (pension funds, hedge funds, proprietary trading desks) face regulatory and contractual leverage limits (UCITS directives, SEC margin requirements, investor redemption triggers). These constraints bind not when funding costs spike briefly, but when they remain elevated for sustained periods, forcing arbitrageurs to reduce leverage persistently and creating selling pressure on the long (low-beta) leg. This is consistent with why the lagged level dominates the contemporaneous change, though the regression cannot rule out omitted variable bias: the shadow price of leverage $\bar{\psi}$ (Section 2.4.1) in Frazzini and Pedersen (2014) reflects aggregate, persistent tightness of funding constraints rather than instantaneous shocks.

Figure 5 illustrates the relationship between the TED spread and rolling BAB returns. The 2008–2009 period, when the TED spread spiked above 4 percentage points, coincides with a sharp decline in BAB performance. The COVID-19 episode in early 2020 offers a second illustration: the TED spread surged briefly, and BAB experienced contemporaneous losses before recovering as funding conditions normalized.

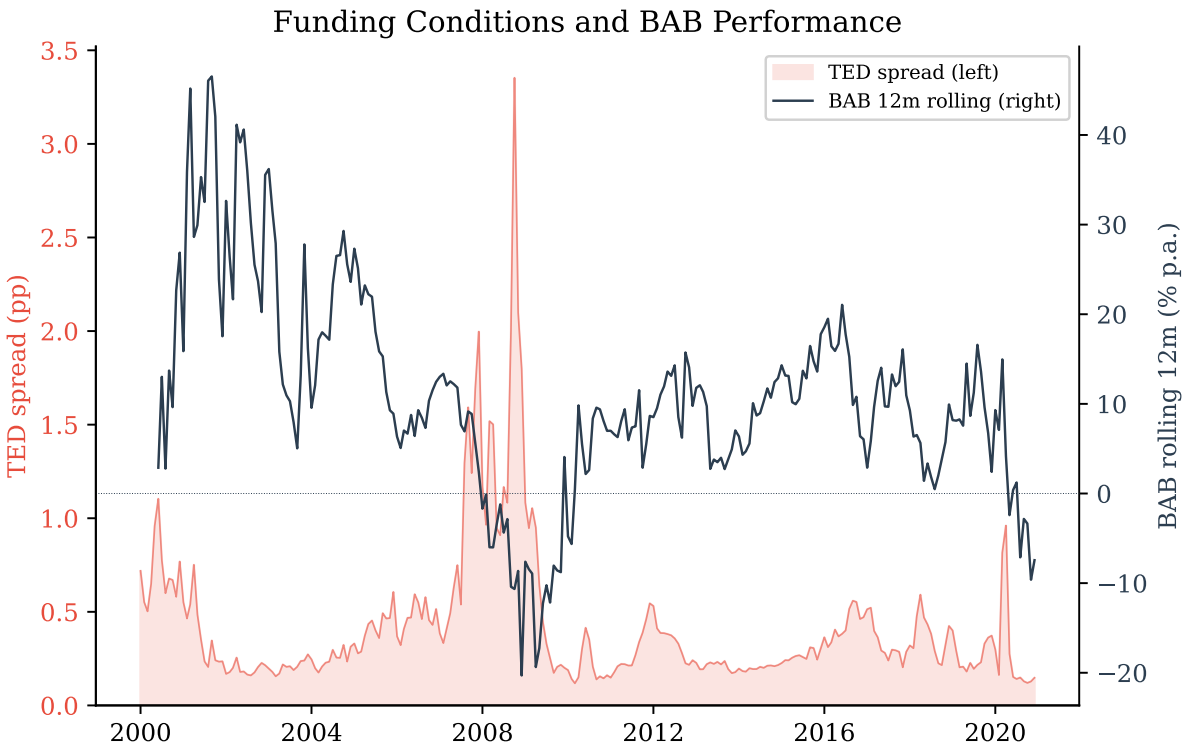

Figure 5: TED Spread and BAB Returns. The figure plots the TED spread (left axis, shaded) alongside rolling 12-month BAB returns (right axis, solid line). Sample: January 2000 to December 2020.

7 Robustness and Limitations

7.1 Robustness Checks

I conduct several robustness checks. First, I verify sensitivity to the winsorization threshold by re-running the entire pipeline without winsorization. Removing winsorization entirely increases the BAB mean return from 10.11% to 10.55% annualized but also increases volatility (from 12.64% to 13.44%), yielding a slightly lower Sharpe ratio (0.785 vs. 0.800). The AQR benchmark correlation weakens from 0.82 to 0.79. Most notably, the D1–D10 alpha spread narrows from 7.52% to 4.71% per annum, driven by D10: without winsorization, the highest-beta decile's CAPM alpha inflates from 0.07% ($t = 0.19$) to 0.60% ($t = 1.90$) per month, reflecting lottery-stock returns that disproportionately affect equal-weighted high-beta portfolios. The BAB factor itself, which uses rank weights, is more robust to outliers. Its FF5 alpha actually increases slightly from 9.45% to 10.03%. These results confirm that winsorization improves data quality and strengthens the decile-level evidence without artificially creating the BAB premium.

Second, I examine beta persistence. Figure 6 displays the month-to-month transition probability matrix for beta quintiles. The diagonal elements range from 86% (Q3) to 96% (Q1), indicating very high persistence: a stock in the highest-beta quintile has a roughly 94% probability of remaining there the following month. This extreme stability reflects the long estimation windows (252 and 1,260 trading days): adding one month of daily data barely changes a five-year rolling correlation estimate. The persistence is necessary for BAB to be implementable; if betas were purely random, the monthly rebalancing required by the strategy would be economically wasteful. It also implies low portfolio turnover, as discussed in Section 7.4.

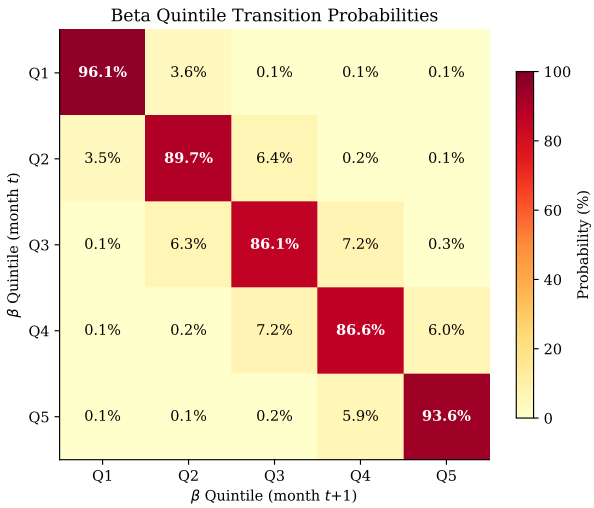

Figure 6: Beta Quintile Transition Probability Matrix. Each cell reports the probability that a stock in quintile i in month t transitions to quintile j in month $t + 1$. Darker shading indicates higher probabilities. Sample: January 2000 to December 2020.

7.2 Methodological Considerations

Decile portfolio returns (Table 1) are computed using equal weights within each decile, because shares outstanding data are not available in the EODHD database. The BAB factor itself (Equations (9)–(10)) uses rank-based weights following Frazzini and Pedersen (2014) and is unaffected by this choice.

Novy-Marx and Velikov (2022, p. 81) demonstrate that rank-based and equal-weighted BAB portfolios produce nearly indistinguishable results, because rank weighting is effectively a "backdoor to equal weighting." Both schemes concentrate portfolio weight in small-capitalization stocks with extreme beta estimates. This has two implications. First, the similarity between the equal-weighted decile analysis and the rank-weighted BAB factor is expected: the weighting choice matters less than the beta sort itself. Second, and more critically, Novy-Marx and Velikov (2022) show that BAB's impressive gross performance is substantially driven by micro- and nano-cap stocks: for every dollar invested in BAB, the strategy commits on average more than one dollar to stocks in the bottom percentile of market capitalization. The persistence of significant BAB alpha under the q-factor and five-factor models, which include profitability and investment factors, provides partial reassurance that the premium is not fully explained by known risk factors, but the small-cap concentration remains a legitimate concern for implementability.

7.3 Alternative Explanation: Coskewness Risk

The leverage constraint hypothesis is not the only explanation for the flat SML, and an important alternative cannot be tested here because it requires option-implied skewness measures constructed from equity option prices, which the EODHD database does not provide. Schneider et al. (2020) propose a risk-based channel rooted in return asymmetry. Their argument starts from a stochastic discount factor that is quadratic in the market return, implying that investors price not only covariance risk (as in the CAPM) but also *coskewness risk*, defined as the covariance between an asset's return and the squared market return. Investors demand compensation for holding assets with negative coskewness, because these assets perform particularly poorly when market volatility spikes.

The connection to the beta anomaly is direct. Low-beta stocks tend to exhibit negative residual coskewness: conditional on their market beta, they co-move disproportionately with large market declines. The CAPM, which prices only linear co-movement, attributes the resulting return premium to alpha. In this interpretation, the positive alphas of low-beta portfolios are not anomalous but reflect fair compensation for a risk dimension that the CAPM ignores.

Empirically, the implications are striking. Schneider et al. (2020) construct coskewness factor-mimicking portfolios from option-implied skewness measures and show that con-

trolling for coskewness renders the BAB alpha statistically insignificant. The CAPM alpha drops from 125 to 33 basis points per month, and the Fama–French three-factor alpha from 109 to 21 basis points. The cross-sectional R^2 between portfolio alphas and residual coskewness betas ranges from 45% to 73%.

This finding does not invalidate the leverage constraint mechanism but suggests that the two channels may operate simultaneously: funding constraints generate excess demand for high-beta assets, while coskewness risk generates a rational premium for low-beta assets. Since my regressions do not control for coskewness, the significant BAB alpha from Section 6.2 is consistent with both interpretations.

7.4 Implementation Frictions

The returns reported here are gross of implementation costs. Several frictions would reduce realized performance:

Transaction costs. Monthly rebalancing generates substantial turnover, particularly as securities cross the median beta threshold. Bid-ask spreads and market impact costs would erode returns.

Shorting costs. The short leg requires borrowing high-beta stocks, which may command significant lending fees, especially for hard-to-borrow securities.

Leverage costs. The BAB construction assumes borrowing at the risk-free rate, an idealization. In practice, leverage carries costs above Treasury rates, precisely the spread that Proposition 3 examines.

The high beta persistence documented in Section 7.1 (86–96% quintile retention) implies that few stocks cross the median each month, limiting side-switching turnover. However, the rank-weighted portfolio also experiences turnover from weight changes as ranks shift and from sample entry and exit. Even modest turnover is costly when applied to the short leg, where hard-to-borrow fees and market impact are substantially higher. Novy-Marx and Velikov (2022, p. 87) estimate average transaction costs of 58 basis points per month for BAB, absorbing more than half of the gross return. After accounting for both transaction costs and factor exposures, they report that the five-factor generalized alpha falls to 17 basis points per month ($t = 1.32$), rendering it statistically insignificant. This is perhaps the most serious challenge to the practical relevance of BAB as a tradeable strategy. These frictions represent the economic costs that limit arbitrage and allow the anomaly to persist.

7.5 Data Limitations

The EODHD sample differs from the CRSP database used by Frazzini and Pedersen (2014). Coverage, adjustments, and treatment of corporate actions vary across providers. The data quality audit identifies 49,383 extreme daily returns (above 100% in absolute value) and 242 monthly observations with absolute returns exceeding 200%, motivating the cross-sectional winsorization described in Section 4.4. The strong correlation (0.82) between my BAB factor and the published AQR benchmark suggests that data source differences do not materially affect the main findings.

8 Conclusion

The introduction asked three questions: why is the empirical Security Market Line flatter than the CAPM implies, which frictions generate this flattening, and whether these distortions can be systematically exploited. All three find answers in the data.

The SML is flatter than the CAPM predicts: CAPM alpha declines from 0.69% per month in the lowest-beta decile to 0.07% in the highest, a spread of 7.52% per annum. The BAB factor earns significant risk-adjusted returns under every factor specification: 12.98% per annum under the CAPM, 13.11% under FF3, 11.78% under Carhart, 9.99% under the q-factor model, and 9.45% under FF5, all significant at the 1% level. And when funding conditions tighten, BAB suffers: the lagged TED spread enters with $t = -3.52$, exactly what the leverage constraint hypothesis would predict.

Why does the beta anomaly persist? The leverage constraint hypothesis provides a clear economic explanation. Many institutional investors face explicit or implicit restrictions on leverage: pension funds are bound by statutory asset-class limits, mutual funds cannot exceed 1:1 leverage under UCITS and the Investment Company Act of 1940, and insurance companies must satisfy risk-based capital requirements. Unable to amplify returns through borrowing, these constrained investors substitute by tilting toward high-beta assets, which offer embedded leverage through greater market exposure per dollar invested. This excess demand compresses high-beta expected returns below the CAPM prediction. The resulting mispricing is not arbitraged away because exploiting it requires precisely the leverage that constrained investors lack. Moreover, as Proposition 3 demonstrates, the strategies designed to capture this mispricing are themselves vulnerable to funding shocks, creating a natural limit to arbitrage that allows the anomaly to persist in equilibrium.

That said, 10% per annum on paper does not mean 10% in practice. Monthly rebalancing generates transaction costs, the short leg requires stock lending fees, and actual borrowing rates exceed the risk-free rate. Novy-Marx and Velikov (2022) estimate that these frictions absorb more than half of the gross BAB premium, rendering the net alpha statistically

insignificant. This is itself a finding: the anomaly persists precisely because the costs of exploiting it are large enough to protect the mispricing from arbitrage. All results are estimated in-sample; the late-sample deterioration documented in Section 6.2 (Sharpe ratio declining from 1.26 to 0.41) and the post-publication decay reported by McLean and Pontiff (2016) suggest that out-of-sample performance would be weaker still. This gap between backtest and implementation probably explains why the anomaly survives despite being well-known.

The replication succeeds in validating the core findings of Frazzini and Pedersen (2014) using an independent data source and a more recent sample period. The correlation of 0.82 with the published AQR BAB factor provides external validation. Several limitations remain: the equal-weighted portfolio construction (necessitated by data constraints), the shorter sample period, and the absence of international markets. Future research could extend this analysis to international equity markets, examine the interaction between BAB and momentum strategies, and investigate the impact of post-2020 market microstructure changes on the beta anomaly.

The flat SML is a robust empirical regularity, documented across samples, countries, and asset classes. The leverage constraint hypothesis explains why it persists, and the data presented here support it. But the gap between gross and net returns is just as important as the alpha itself: the anomaly survives because exploiting it is expensive enough to keep the mispricing in place. A statistically significant alpha is not the same as a profitable trading strategy.

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